

StaND4LQ

State of the Art & New Directions for
Lattice Gauge Theory and Quantum Simulation

Polo Fibonacci · Pisa · 30 September – 2 October 2026

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University of Pisa · organized by PhD students, supported by INFN Sezione di Pisa



Benvenuti a Pisa — and welcome to the **first edition** of StaND4LQ.

This workshop is organised **entirely by PhD students** of the University of Pisa.

No senior committee, no registration fee — just the meeting we wanted to attend.

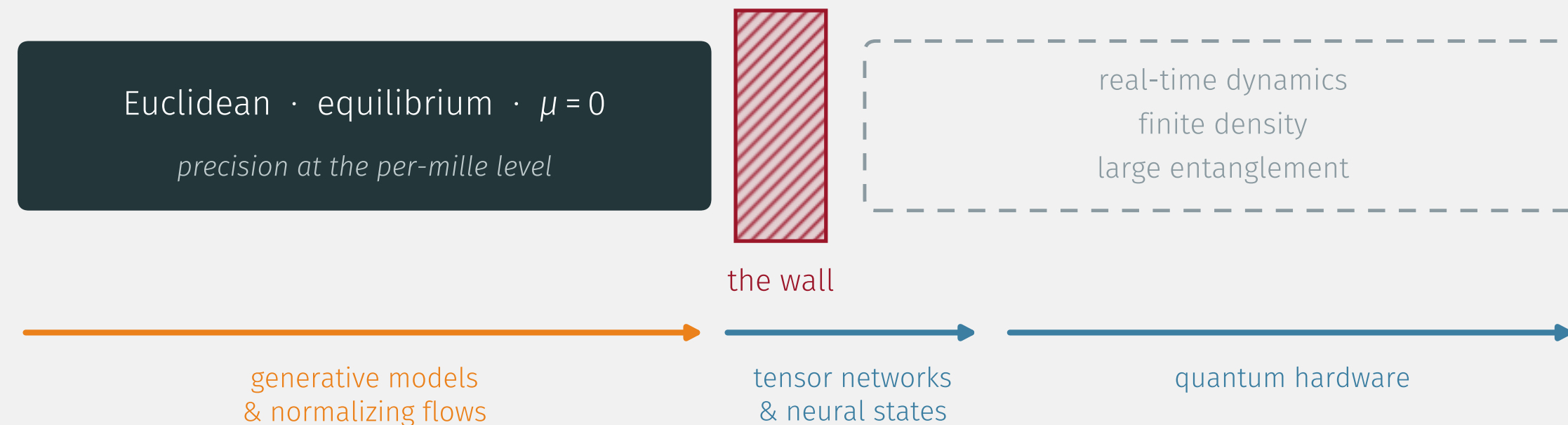
Three days, **eighteen talks**, **ten invited speakers**, and one long discussion with no talks at all.

Lattice field theory is now a precision science

...and precisely *because* it is, we can see clearly where it stops.

STATE OF THE ART

NEW DIRECTIONS



The wall is not a technicality. It is a **sign problem**, an ill-posed **analytic continuation**, and a Hilbert space that grows **exponentially**.

What blocks us

- efficient **Monte Carlo** sampling
- **real-time** dynamics
- systems at **finite density**
- gauge theories on **quantum devices**

Who is needed

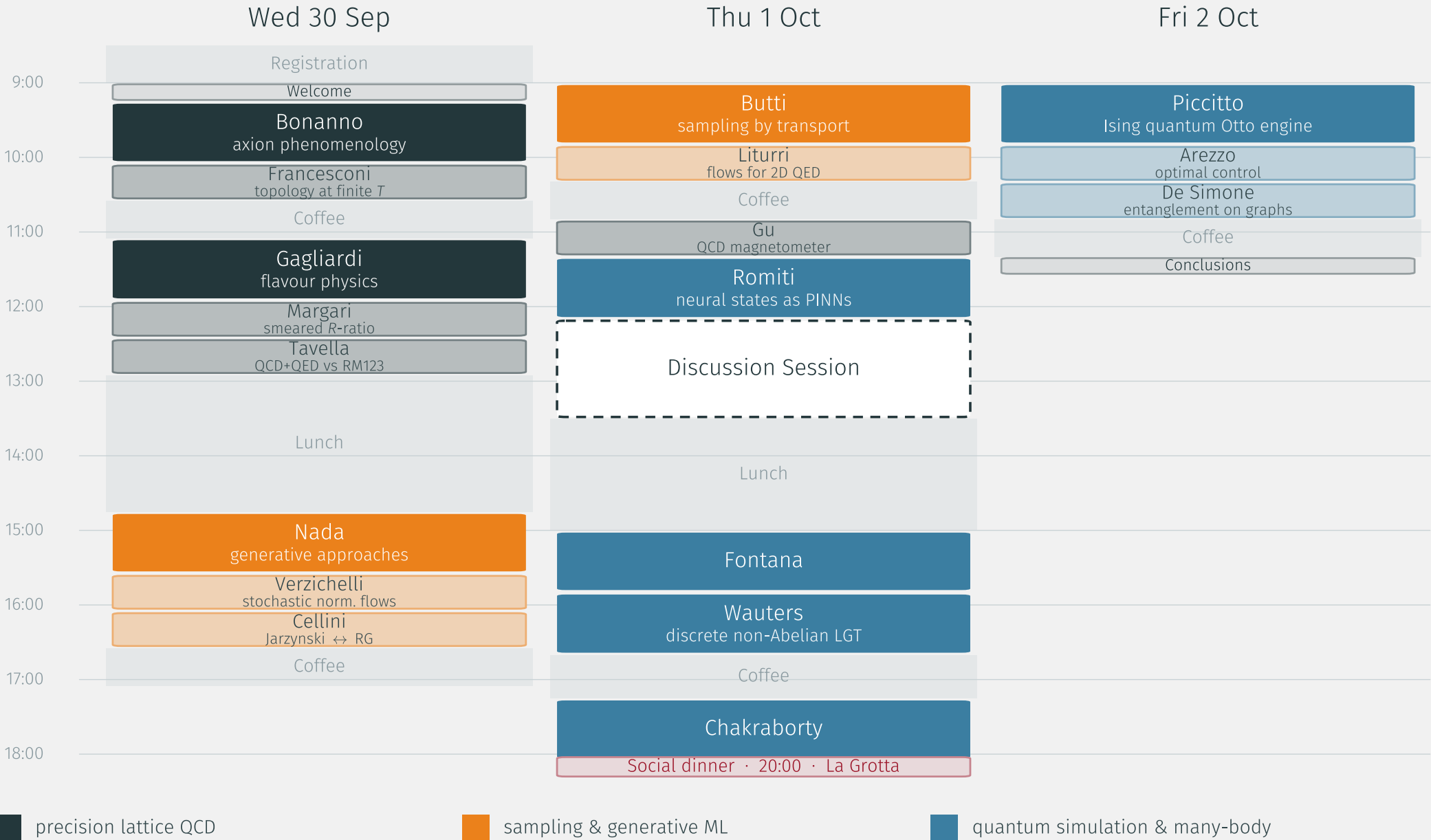
- lattice **field theory**
- **quantum information**
- **machine learning**
- quantum **many-body** physics

None of these four problems belongs to a single community —
and that is the entire premise of this meeting.

The programme is the argument



Read it left to right: **state of the art** on Wednesday, **new directions** by Friday.



Ten invited speakers



UNIVERSITÀ
DI PISA



Claudio Bonanno

University of Bern

Elia Cellini

University of Edinburgh · Higgs Centre

Pierpaolo Fontana

Universitat Autònoma de Barcelona

Alessandro Nada

Università di Torino · INFN Torino

Simone Romiti

University of Bern

Pietro Butti

University of Southern Denmark

Bipasha Chakraborty

University of Southampton

Giuseppe Gagliardi

INFN, Sezione di Roma Tre

Giulia Piccitto

University of Catania

Matteo Wauters

University of Trento

Nine contributed talks from early-career researchers sit alongside them — by design, not as filler.

A first edition, in numbers

31

registered participants

18

talks in three days

5

countries represented

0

registration fee

Italy · United Kingdom · Switzerland · Denmark · Spain

For a first edition, assembled by four PhD students, we are quietly delighted.

Thursday 1 October · 12:10 – 13:30

An hour and twenty minutes with **no speaker and no slides**.

It is the only part of the programme we cannot prepare for you.

If you came here to find a collaborator rather than to give a talk,
this is the slot that matters.

Venue

Polo Fibonacci, Building A
Aula Magna, Dept. of Mathematics
Largo Bruno Pontecorvo 3, Pisa

Coffee breaks are offered by the organisation, just outside this room.

Social dinner

Ristorante **La Grotta**, Pisa
Thursday 1 October, **20:00**

Anything at all — find any of the four of us, or write to
stand4lq_workshop@lists.infn.it

This workshop is supported by the **University of Pisa** and **INFN, Sezione di Pisa** — which is why nobody paid to walk through the door.

Thanks to our **invited speakers** for agreeing to come to a workshop that did not yet exist, and to everyone who **submitted an abstract**.

One request · the people in this room work on Euclidean Monte Carlo, on tensor networks, and on quantum hardware. Sit next to someone whose methods you do not use.

Enjoy the workshop!